



OCTAVIA CARBON

Carbon dioxide removal prepurchase application 2025

General Application

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Octavia Carbon Co.

Company or organization location (we welcome applicants from anywhere in the world)

Nairobi, Kenya

Name(s) of primary point(s) of contact for this application

Duncan Kariuki

Brief company or organization description <20 words

Octavia Carbon develops solid sorbent direct air capture that integrates Kenya's geothermal heat to cover most of its energy requirements.

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

We develop solid sorbent Direct Air Capture (DAC) technology that integrates with Kenya's abundant

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

geothermal heat to supply the majority of its energy needs, while permanently storing CO₂ through in-situ mineralisation through our storage partner, Cella Mineral Storage. As the Global South's first DAC company, we have understood that DAC will only replicate solar's cost curve if we also replicate a value chain efficiency that rapidly scales insights from the lab into low-cost manufacturing & a project development that isn't resource-constrained by energy or storage. Kenya's unique combination of renewable energy resources, geological storage capacity, and skilled workforce provides us with the ideal conditions to scale DAC, and our proprietary technology is designed to fully leverage these advantages. This combination of location, technology, and execution positions us to scale rapidly while delivering on Frontier's prepurchase priority innovation areas: *Efficient and flexible energy use*, and accelerating *in-situ mineralization*.

At the heart of our technology is a solid sorbent that is designed for fast cycle times, high volumetric capacity, and rapid tunability, allowing us to achieve high capture capacities at low cost. The sorbent is housed in a cylindrical pressure vessel ("the reactor"/"the DAC unit"), optimized for low-cost manufacturing and efficient real-world operation. Multiple reactors are linked through a shared balance-of-plant that supplies steam, vacuum generation, cooling water, dehydration, and liquefaction.

The CO₂ capture process operates in two steps: adsorption and desorption. In the adsorption phase, ambient air is drawn through the sorbent, which selectively binds CO₂ while allowing other gases to pass through. Breakthrough analysis continuously monitors the amount of CO₂ adsorbed by the sorbent. The system transitions to desorption once the sorbent hits 95% saturation. During desorption, the reactor is sealed and a partial vacuum is applied. Non-condensable gases such as nitrogen, oxygen, and inert gases are purged out of the system using steam, creating optimal conditions for desorption. The sorbent is then heated up to >80°C under a mild vacuum, releasing the bound CO₂. The resulting gas stream is dried yielding CO₂ at >99% purity. This is then liquefied and stored in insulated cryogenic tanks, awaiting storage. Once desorption is complete, the sorbent is cooled to its optimal adsorption temperature, and the capture cycle begins again.

Following the DAC process, we have partnered with Cella Mineral Storage ('Cella'), an in-situ carbon mineralisation company operating in Kenya and part of Frontier's Winter '22 cohort, to provide CO₂ storage. We are currently trucking CO₂ to Cella because our pilot is built around existing grid connectivity a few miles away on the same site. From the full-scale Project Hummingbird onward, we will co-locate with Cella and pipe CO₂ directly to their storage facility.

The CO₂ is injected into porous basaltic rock formations, where it reacts with calcium and magnesium silicates to form stable carbonate minerals. The Kenyan Rift Valley's geology is ideal for this process, enabling rapid and complete mineralisation. Once converted into carbonates, the CO₂ remains locked away for millions of years.

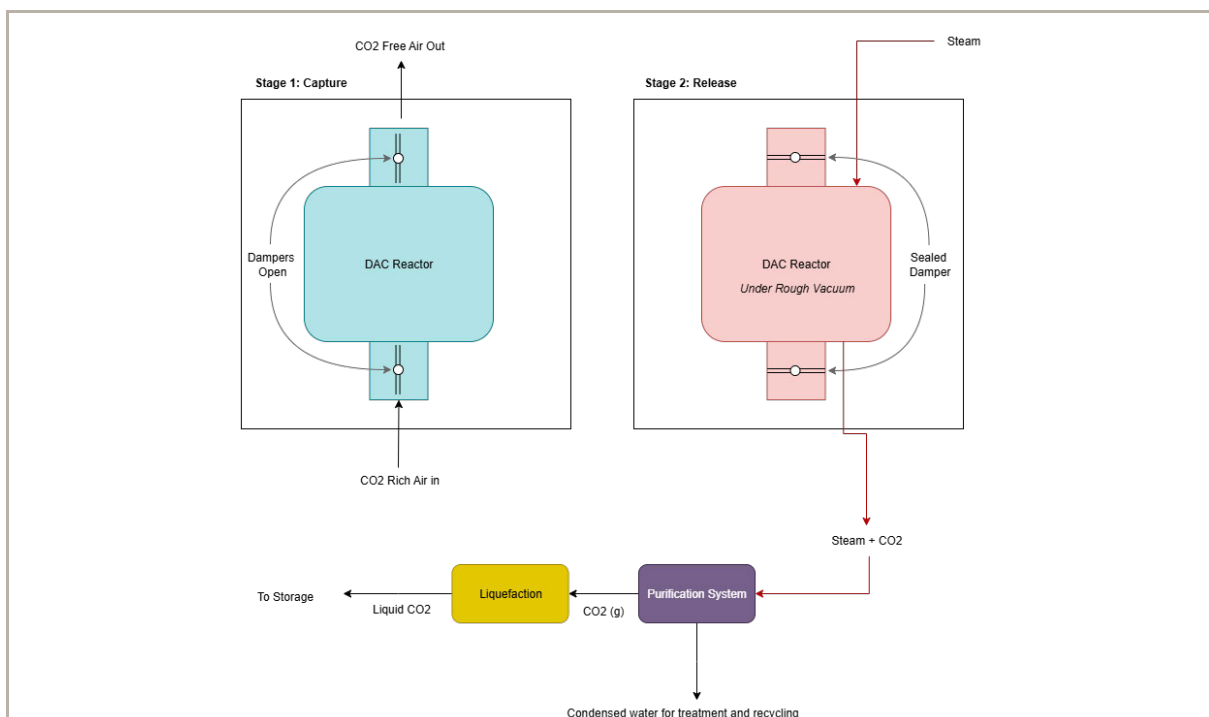


Figure 1: System schematics

Our approach is designed to overcome the structural barriers that have kept DAC expensive and slow to scale. We are harnessing Kenya's unique endowments, renewable energy, favourable geology, and skilled talent, to unlock a fundamentally different cost and scale trajectory.

Our deployment site hosts the world's third CO₂ mineralisation well alongside an active geothermal concession licensed for 140 MWe of electricity and over 1 GW_t of heat. This co-location of energy and storage is exceptionally rare and provides us with a structural advantage that few other DAC companies can access. From the outset, we opted to design our machines to leverage low-grade heat, ensuring compatibility with the abundant geothermal resources, ultra-low-cost, and fully renewable energy source. In 2026, we will begin drawing steam directly from dedicated well heads, which will supply the heat required for our operations. As of today, we have achieved >4,000 operational hours at this site, producing storage-grade CO₂ that Cella will start injecting soon.

Our systems utilise the low-grade geothermal steam for 80–85% of their energy needs, with the remaining 15–20% supplied by geothermal electricity for fans, pumps, and auxiliary systems. In Kenya's Rift Valley, this heat is accessible at shallow depths (~1,000 ft) and at ultra-low cost (\$0.01–\$0.02/kWh), enabling continuous, fully renewable DAC operations with minimal operating costs and carbon footprint. We have designed our technology to efficiently integrate with this low-cost heat source, from extracting heat effectively from steam to recovering heat downstream of our reactors. With support from the Bezos Earth Fund, we are enhancing our sorbent materials to better withstand exposure to hydrogen sulfide (H₂S), a common impurity in geothermal environments. This will improve sorbent durability and efficiency, enabling more reliable long-term integration with geothermal energy.

Our speed of innovation and modular approach have cut down our DAC development timelines to

months, slashed costs between builds, and enabled rapid, scalable deployment without sacrificing performance. At plant level, we have developed a plant engineering design that maintains a high level of modularity, allowing us to scale the construction of our plants rapidly, while maintaining a high learning rate between plants. We have built the EPC capacity in-house to give us tight control over the EPC process of the plant. This strategy paid off in the construction of our field pilot plant which went from design to commissioning in <6months, and was built both on time and on budget. For our recently deployed Gen2 tech field pilot, we managed to shrink the iteration time to 5 months. Over this window, our performance increased 5x, and capex reduced 50% on a levelized basis. This speed of innovation is driven by local manufacturing, resilient short supply chains, advanced simulation and AI tools, and a highly skilled Kenyan team.

At the technology level, we have formulated a sorbent from earth-abundant materials that offers rapid cycling, high volumetric capacity per cycle, and extremely high tunability. This tunability enables us to continually enhance our sorbent's performance through microscopic adjustments to the wash coat layer. This would enable us to improve the capacity and energy efficiency of our materials, such as by incorporating Layered Double Hydroxides, which we are studying, without changing our reactor design. In addition to refining the sorbent formulation, we have established a low-cost supply chain with global chemical manufacturers, providing an affordable pathway for scaling up production.

The sorbent is housed in a reactor designed for low-cost, large-scale manufacturing. We designed our reactors as cylindrical pressure vessels that use 40% less steel than traditional DAC reactor designs. We are uniquely capable of building this form factor because we have developed strong in-house manufacturing capacity, leveraging our low-cost manufacturing base in Kenya. The modularity of our reactors has also helped us learn quickly, enabling us to significantly reduce costs and improve manufacturing efficiency. For instance, from the first to the fourth reactor in our initial field pilot plant, we cut the reactor building cost by 50% and reduced manufacturing time by 88% within just six months. We also focus heavily on minimizing the CAPEX of the DAC reactors and the plant, continuously refining and streamlining our designs to build faster and more affordably. For example, our second-generation reactor has 520 parts, compared to 4,340 parts in the first-generation reactor.



Figure 2: Nelion unit (our Gen 2 tech) with the initial 2 submodules deployed

Our approach aligns closely with Frontier's DAC priority innovation areas. Leveraging low-grade geothermal steam for 80–85% of our energy needs ensures highly efficient, fully renewable operations

with minimal carbon intensity. East Africa's Rift Valley holds one of the world's largest untapped geothermal resources, with Kenya alone ranking as the 6th-largest geothermal power producer globally and the largest in Africa. Kenya's geothermal sector is expanding rapidly, and with it, significant amounts of low-grade heat will remain underutilized in the coming decade. Studies indicate that 50–70% of geothermal fluid energy is typically not converted to electricity², creating a long-term opportunity to power DAC at very low cost³. Our pilot plant will use low-grade steam from a dedicated geothermal heat well, and as we scale, we will explore integration with geothermal power plants to drive down our costs further. By leveraging geothermal heat at costs as low as ~\$0.02/kWh_t, we can supply the bulk of the energy needs for our DAC at minimal cost while eliminating a major source of wasted renewable heat.

Lastly, through Cella, the captured CO₂ is permanently stored via in-situ mineralization, where it reacts with basaltic rocks to form stable carbonates, ensuring durability for >1,000 years. Together, we are on track to be among the first globally to achieve permanent atmospheric CO₂ storage via DAC and geological injection, reaching this milestone in ~3 years of operations and with about \$10 million in funding between Cella and us.

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?⁴ What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

Introduction

Octavia Carbon, in partnership with Cella Mineral Storage, is developing Project Hummingbird, the Global South's first DAC facility. Located in Kenya's Rift Valley within the Sleeping Warrior Special Economic Zone in Gilgil, the project will remove and permanently sequester more than 1,000 tonnes of atmospheric CO₂ each year through geological storage.

We strategically chose this location for its exceptional geological storage capacity and abundant geothermal energy, which together provide the two essential prerequisites for scaling DAC in one site. With a nameplate capacity of 1,500 tonnes of CO₂ per year, the facility will deliver roughly 10,000 net verified removals over its lifetime, after accounting for embodied emissions and MRV buffers. We have planned the project for 10 years of full-scale operations.

Project Hummingbird marks the first commercial-scale deployment of Octavia's geothermal-integrated

² DiPippo, Ronald. Geothermal power plants: principles, applications, case studies and environmental impact. Butterworth-Heinemann, 2012.

³ Frick, Stephanie, Martin Kaltschmitt, and Gerd Schröder. "Life cycle assessment of geothermal binary power plants using enhanced low-temperature reservoirs." Energy 35, no. 5 (2010): 2281-2294.

⁴ We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

DAC approach, paired with Cella’s in-situ CO₂ mineralisation in basalt formations. We will power the project with 100% clean, firm energy—drawing initially from Kenya’s renewable-rich grid, supplemented by solar and batteries, and later from an onsite geothermal power plant supplying power behind the meter. A shallow geothermal heat well onsite will also provide the project with clean, firm heat.

Development of the project will follow a phased approach to reduce risk and maximise learning rates between phases of the project. Phase 0, a 30tCO₂/year pilot, has been operational since early this year, with >4,000 cumulative operational hours to date. This phase has validated our process through the deployment of four Gen 1 reactors alongside our first Gen 2 reactors. It has provided critical validation of costs, performance metrics, and operations & maintenance (O&M) requirements.

Costs:

Project Hummingbird has a cost profile of <\$626/tCO₂ (unsubsidized) for the capture and permanent sequestration of CO₂ over a 10-year plant lifetime. Our technology faces a steep cost decline trajectory, primarily driven by modularity, scale, and advancement of our technology. These factors will drive the cost of our DAC down to meet Frontier’s cost and scale targets.

Our DAC units are modular vessels, allowing a steep learning curve between each build. For example, phase 0 of our pilot plant had 4 DAC units of our gen 1 design. We achieved a 50% cost reduction in the CAPEX per unit between June & December 2024 as we constructed and commissioned the first phase of our pilot plant. Simple iterative design improvements and operational efficiency gains drove this steep cost reduction. Equally, as we have advanced to a 2nd-generation technology platform, we have improved the design of our DAC units, by cutting out >90% of the number of parts in each DAC unit, which has simplified the manufacturing and reduced cost of the unit. We expect this learning rate to continue as we build more DAC units.

Equally, further cost reductions will arise from advancement in our technology. This is everything from improved sorbent performance (higher productivity), improving cycle times, improving energy efficiency, integrating heat energy recovery etc. In 2025 alone, we have increased the daily volumetric productivity of our sorbent by >12X, primarily by increasing the amount of CO₂ we desorb in every cycle and increasing the number of cycles. We continue to innovate on our materials and process engineering to increase this productivity further.

The 3rd lever that will drive down our cost is scale. This will manifest particularly in the balance of plant equipment like our liquefaction plant, vacuum pumps, heat exchangers etc. that enjoy favorable economies of scale on energy and unit costs. We have also worked to develop a low-cost supply chain for our inputs that spans China, India, Germany, and Kenya. As we scale our procurement of raw materials, we will further optimise this supply chain and drive down our unit costs.

Table 1 shows our cost reduction expectation as we scale from our pilot to our FOAK commercial facility, achieving close to 60% cost reduction in the cost of removal. We expect this trend to continue as we scale. *[Detailed breakdown of the costs available in the attached TEA]*

Table 1 Cost breakdown as we scale

	Cost Driver	1500 tCO ₂ /yr		10,000 tCO ₂ /yr	
		\$/tCO ₂		\$/tCO ₂	
C A P E X	DAC reactors	60	358	16	120
	Plant Equipment	78		21	
	Financing	156		68	
	Contingencies	49		12	
	Engineering & Permitting Costs	15		3	
O P E X	Levelized Energy Cost	90	290	42	132
	Other Variable OPEX (Admin, Plant Overheads)	30		11	
	Fixed OPEX (Labour)	62		20	
	CO ₂ transport and storage Costs	93		50	
	MRV: Audit & Registry fee	15		9	
	Total Cost of Capture (\$/tCO ₂)	648		252	

Scale:

Our uniquely deep vertical integration gives us an edge in delivering against Frontier's scale-up criteria. To achieve this, we have developed a highly modular technology from the sorbent structure design to the design of the DAC unit to the plant itself.

We use a highly modular sorbent structure, which allows high tunability and cost-effective scale-up. At the time of writing, we are stress-testing a new sorbent formulation with 2.2× higher productivity than the one currently capturing CO₂ in our field pilot. We plan to retrofit this formulation into existing DAC units, boosting daily productivity without altering the plant infrastructure.

Our DAC units follow the same modular principle. Each container module houses four submodules (see Fig. 1). Between the first and second modules, we project a 30% CAPEX reduction, even while increasing daily CO₂ capture productivity by 2–3×.

We have also designed and built our plant to be modular, by skid mounting all the equipment in the balance of plant from the liquefaction plant, cooling tower, vacuum and dehydration systems. This allows fast build times which engrain strong learning rates at project level. We built our first plant in 6 months from design to vendor engagement and procurement and final commissioning. This allowed us to identify where lead time constraints occur with plant equipment and strategies for mitigating long lead times and high costs. The standardization allows efficient scale-up and economies of scale, where we are projecting >24% cost reduction for every doubling of plant capacity.

Moreover, we are deploying our technology in Kenya's rift valley, which has abundant and untapped clean power and heat. As such, we don't expect that energy will be a constraint to the scale-up of our technology.

MRV Strategy

We quantify CO₂ removal by the DAC plant across four dimensions: captured CO₂, leakage, embodied emissions, and operational emissions. For captured CO₂, we monitor the mass at each stage of adsorption, desorption, and liquefaction. We deploy a suite of instruments to track pressure, temperature, mass flow rates, and purity levels, applying thermodynamic relations to calculate the exact mass of CO₂ captured. At custody transfer, we verify and report the CO₂ mass delivered for storage using both a load cell, which logs changes in mass during transfer between cryogenic tanks, and a mass flowmeter, which measures the flow rate in the transfer pipe.

To ensure accuracy, all sensors are regularly recalibrated and maintained, with data stored securely in the cloud for transparent reporting. A maintenance database tracks sensor performance and schedules recalibration. These data points are centralized through a SCADA system, enabling remote monitoring, continuous performance tracking, and automated alerts if parameters deviate from thresholds. In parallel, we are co-developing the first digital MRV system tailored for DAC with Carbonfuture to further enhance transparency.

For embodied CO₂, we partnered with a third party to conduct a cradle-to-grave Life Cycle Assessment (LCA) in accordance with ISO 14040 and ISO 14044. We work with our vendors and rely on government data and reputable databases such as Idemat and Ecoinvent to monitor embodied and operational emissions. Our credits are issued under Puro.earth’s GSC standard, verified by DNV.

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

Technical Risk: Shorter Plant Lifetime

We have designed the pilot plant to run for 10 years, and its CAPEX and carbon footprint are amortised over a 10-year plant lifetime, feeding our unit economics. There is a risk of shutting the plant down earlier than its expected lifetime, as we have seen with other pilot plants in the industry and in adjacent industries. This could risk future delivery and raise the cost of any credits delivered over the lifetime of the plant, while raising the embodied emissions of any delivered credits.

To mitigate this, we are reserving our plant’s excess removal capacity in the initial years of operation for such contingencies. While we expect to remove 1150 tCO₂/yr, we will only initially issue 1000 tCO₂/yr to customers. Additionally, we are designing our technology to be highly tunable and modular, allowing us to retrofit our project with higher-performing sorbent and DAC units in the future.

Technical Risk: High Sorbent Degradation Rate

Our sorbent has 2 main degradation mechanisms: oxidative degradation and amine leaching. These impact the sorbent’s productivity over time, reducing productivity and raising material costs. We mitigate oxidative degradation by running desorption in rough vacuum and in the absence of oxygen. We mitigate leaching by using non-condensing steam, which has resulted in very low cyclic leach

rates from our Gen2 field pilot, validating our conservative TEA assumptions.

Project Execution Risk: Overruns

Like any other infrastructure project, our DAC facility faces the risk of cost and timeline overruns. Procurement delays, utility integration, and permitting are the main risks. We are phasing construction to reduce exposure so each stage de-risks the next. This approach shortens learning curves, builds vendor relationships, and reduces lead-time uncertainty. We are also oversizing key utilities upfront; installing a transformer that can power the full Project Hummingbird, though we will initially use <10% of its capacity. By phasing buildout, we bring regulators along with us, helping them gain familiarity and build capacity for permitting DAC projects.

Energy Risk: High Cost of Energy

Kenya's energy is clean with a very low carbon footprint, but grid electricity is quite costly at over \$170 per MWh, which would raise our levelized cost of capture. We plan to source power directly from an onsite geothermal plant co-located with our DAC and storage well to address this. This would offer benefits in cost, emissions certainty, and energy reliability. Currently, the geothermal plant is still in the early stages of development. We will rely on the grid for electricity before the onsite geothermal power plant comes online.

To mitigate this risk, we are in the process of securing a Special Economic Zone License that will allow us to procure power at a special tariff rate of \$110/MWh from the grid and reduce our demand for grid electricity by integrating our own captive use solar plant, while relying on lower-cost off-peak grid tariffs at night.

In addition to electricity, our plan is to also use geothermal heat in the form of steam from a shallow dedicated heat well. This dedicated heat well has not yet been drilled, which could delay the supply of geothermal heat to the project. Fortunately, this well has been de-risked by a grant from the African Union that will appraise the resource, plus Kenya has seasoned developers with decades of experience in geothermal drilling.

Financial Risks:

The cost of capital for infrastructure projects is significantly higher in Kenya and much of Africa compared to the Global North, risking our future cost reduction. We will finance Project Hummingbird entirely from our Series A equity funding as well as customer prepayments. In subsequent projects, we will begin to layer in lower-cost debt into our capital stack.

To mitigate this risk for future projects, we are building strong relationships with development finance institutions (DFIs) and impact investors that can provide concessional capital at rates closer to those seen in Europe and North America.

Credit Issuance Risks:

Issuing DACCS credits is inherently challenging, with only one company having done so to-date. The barriers lie in demonstrating both technological readiness and a robust, science-backed MRV plan. We have de-risked the technology to TRL 7 and deliberately chosen to work with proven leaders i.e.

Puro.earth and DNV to ensure our credits meet the highest verification and quality benchmarks. Given that this is a FOAK project, timelines remain a key risk. To mitigate this, we are conducting a preliminary assessment with Puro to validate our MRV plan and accelerate delivery. Additionally, we engaged independent rating agencies (Silvera and BeZero) to evaluate our project, surface risks, and refine our approach to strengthen credit issuance readiness.

Permitting Risks:

DAC is still a new technology in Kenya (and everywhere), and authorities like the National Environment Management Authority (NEMA) do not yet have specific reviewer expertise for these types of projects. This often results in multiple feedback cycles as these regulators seek clarification on novel aspects of our operations, which can lengthen approval timelines. To mitigate this, we are closely engaging with the relevant authorities to clarify permitting requirements and in the preparation of the required documentation. This is key to reducing the risk of protracted delays on our operations and helps the regulators build familiarity with DAC projects.

Storage Risk

CO₂ sequestration is central to our carbon credit delivery, and reliance on a storage partner introduces counterparty risk. To address this, we have partnered with Frontier-recognized storage developer Cella, led by a seasoned geoscience team, leveraging Kenya's world-class basalt formations for CO₂ mineralization. Their credible expertise & strong project development traction gives us confidence that storage will remain a reliable component of our delivery pathway.

- d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	714 tons
Delivery window <i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	2026- 2028
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	\$648/t
Levelized price (\$/ton CO ₂) ⁵ <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	\$700

⁵ This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).